

AMCT-Pruning 结构化剪枝技术

Three Scopes on Pruning / 剪枝三个维度的视角

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Agenda

01

Why pruning – What is it, What is the cost

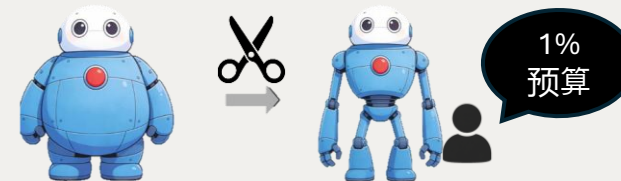
为什么剪枝



02

The tool – CANN/AMCT pruning

剪枝工具



03

What's next – Ongoing Researches

下一步



Why pruning

为什么剪枝

FNN
CNN
MoE



• Example: LLaDA2.1-flash · 100 GB

36 GB over

does not fit at all

910B3 NPU 单卡 · 64 GB

what one accelerator holds

A 100GB model does not fit on a 64GB accelerator. What can you remove, and what does removing it cost?

100GB的模型放不进 64GB 的加速器。模型是否能裁剪？代价是什么？

IN THIS CHAPTER

- What structured pruning actually cuts
- Where the interfaces have to line up

→ You cannot serve what you cannot fit. / 装不下，就谈不上部署。

Structured pruning : three axes

为什么剪枝 · 04

Prune only consumer can take it. Blue marks removed structure.

MECHANISM · 机制

DENSE FFN

intermediate dim

W1 2048 → 8192

W2 8192 → 2048



shared hidden axis

DENSE_LOWVAR 50%

CNN

output channels

Conv2d out

BatchNorm

next Conv in



3 tensors together

CNN_RECONSTRUCT 30%

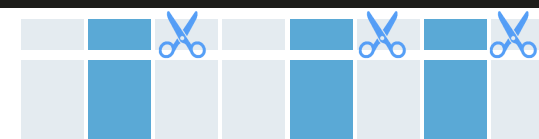
MOE

whole experts

router

experts

expert + router row



MOE_MASSVAR 50%

FLOW · 流程

SCORE

e.g.) variance · routed
mass · reconstruction

PRUNE

slice weights and the
consumer axis, in place

RECOVER

finetune pass · lr 1e-5

loss ≤ tolerance?

the search decides, not
the caller

QUANTIZE

a separate axis, under the
same budget

over budget → try a smaller ratio

one pass of amct.prune + amct.quantize

HEADLINE · 关键结果

0% → 27%

Dense-LM

FFN · 1% perplexity budget · one finetune pass

0% → 17%

granite-1b-a400m

MoE · 1% perplexity budget · one finetune pass

0% → 30%

MNIST CNN

CNN · output channels · reconstruct criterion

→ Experiments demonstrated, under a 1% performance degradation threshold, an average of 27%, 17%, and 30% pruning could be achieved for FFN, MoE, and CNN (maximum 61% for FFN). 实验表明, 在**性能下降不超过1%**的条件下, FFN、MoE 和 CNN 分别可实现平均 27%、17% 和 30% 的剪枝率, 其中 FFN 的最高剪枝率可达 61%。

The tool

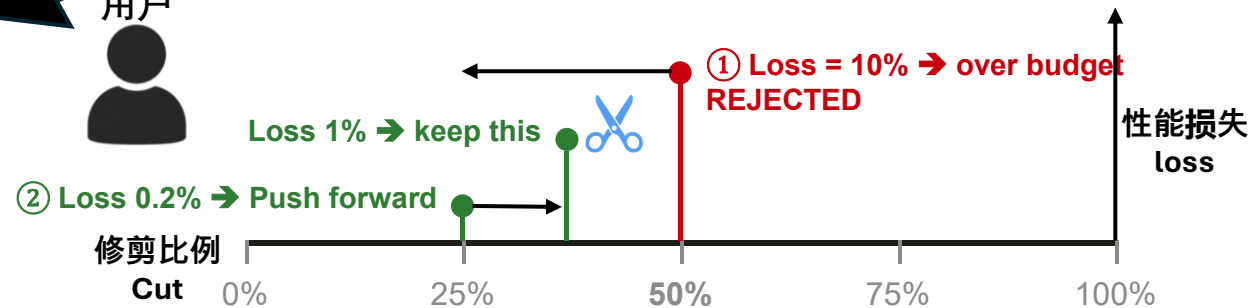
剪枝工具

Tolerance = 1%

1% ✂ 预算



- one binary search over the ratio grid



Nobody knows the right ratio in advance. So do not ask for a ratio — ask for the loss you can accept.

正确的配比无法预先确知，故不应聚焦于比例本身，而应明确界定可承受的损失限度。

IN THIS CHAPTER

- You state a tolerance, the tool finds the ratio
- Recovery is a menu, and a guard makes the pick
- Where the scoring criterion helps, and where it does not
- One tolerance across pruning and quantization
- What it costs to adopt

→ User-friendly interface designed for ease of use./ 该界面秉持用户友好设计理念，致力于实现操作便捷性与易用性

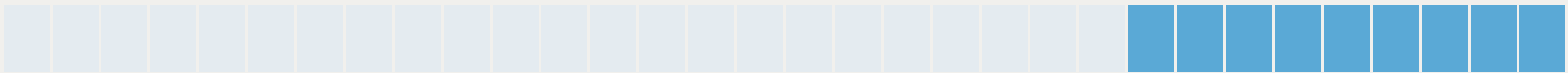
Tolerance search : loss in, ratio out

The caller supplies only the accuracy loss they can accept.

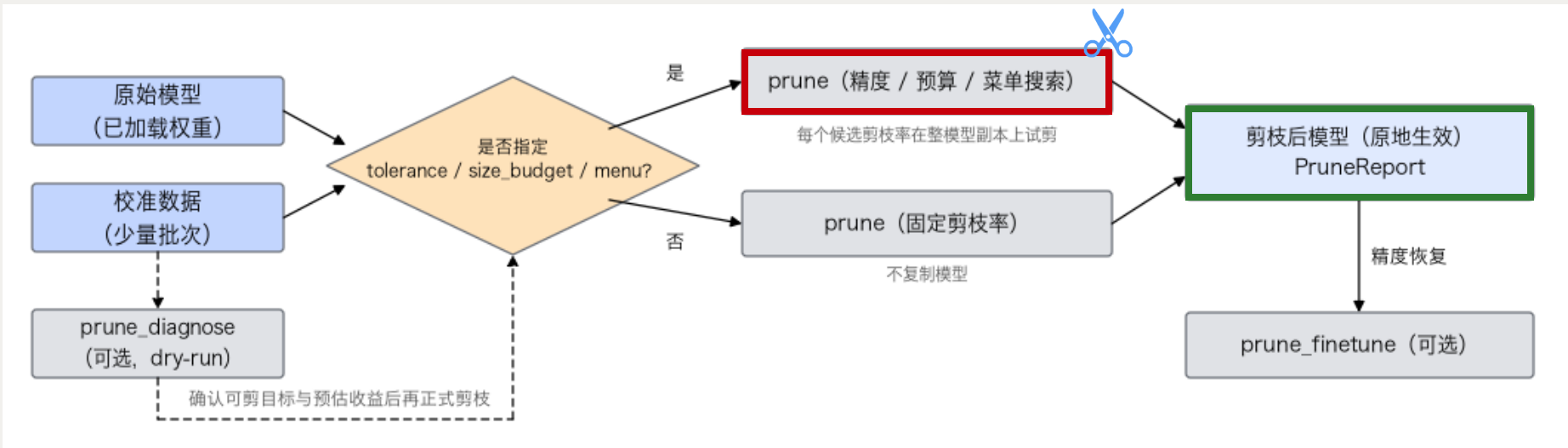
THE MODEL · 模型如何变化

dense FFN intermediate dim

the largest ratio that still meets the budget



CALL FLOW · 调用流程



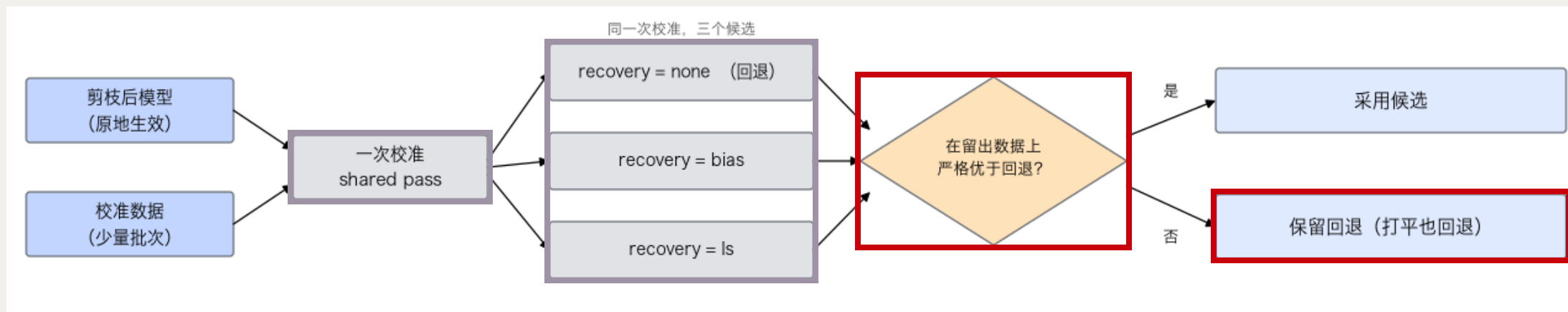
→ Through an automated pruning process, the system identifies the optimal pruning ratio based on the user-defined accuracy-loss threshold, enabling safe, high-performance pruning without requiring any prior knowledge. / 借助自动化剪枝机制，系统依据用户设定的精度损失阈值，自动识别最优剪枝比例，从而在无需先验知识的前提下，实现兼具安全性与高性能的模型剪枝。

Recovery menu : the held-out guard

剪枝工具 · 07

Pruning removes structure. Recovery decides whether the model comes back

CALL FLOW · 调用流程



EVIDENCE · 证据

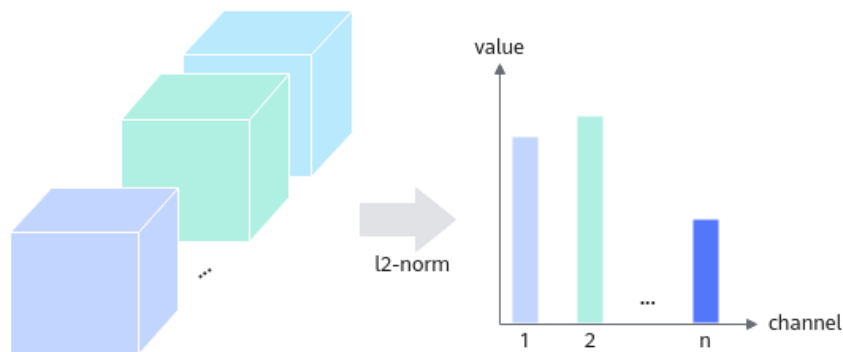
Entry	Domain	What it does	When it is adopted
recovery = none	dense FFN, CNN	slice and keep going	the fallback; also the tie-breaker
recovery = bias	dense FFN, CNN	folds removed units into the bias	strictly beats the fallback
recovery = ls	dense FFN, CNN	least-squares reconstruction of the output	strictly beats the fallback
moe = mass / cond_var / peak	MoE experts	the expert-scoring menu, same rule	same guard, same held-out split
finetune_fn	any domain	a real finetune pass inside the search	when training data is supplied
guard = calib_nll	cross-layer allocation	reverts when calibration degrades	always on

→ Nothing is adopted because it looked better on the calibration set. A candidate ships only if it strictly beats the safe fallback on held-out data, and ties go to the fallback. / 任何候选方案均不得仅因在校准集上表现更优而获采纳。唯有当候选方案在保留数据集上的表现严格优于安全备选方案时，方予发布；若两者表现相当，则维持采用安全备选方案。

CNN scoring criteria : the ranking decides

Thirty percent of a CNN's channels can go with no training step of any kind, and the accuracy does not move.

SCORING · 评分准则



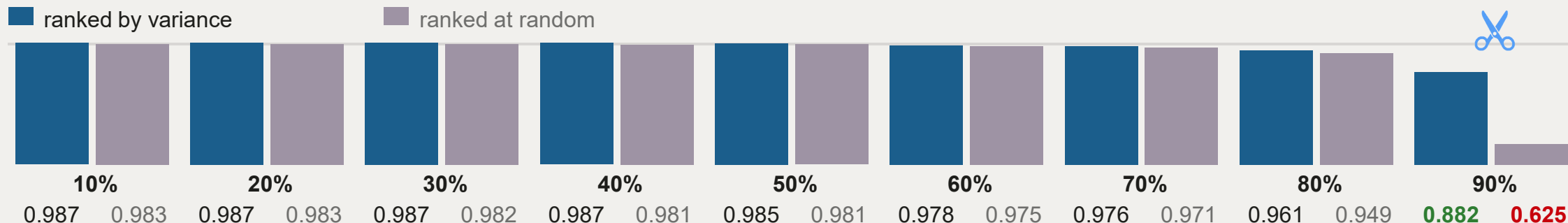
REMOVE 30% OF THE CHANNELS

0.9869 → 0.9871

no gradient step
after the cut

no finetuning, no recovery pass, no calibration beyond the 512 images used to rank
and the ranking is worth almost nothing here — the model has the channels to spare
at 90% it decides everything: 0.8823 ranked against 0.6245 at random

MEASURED · 实测

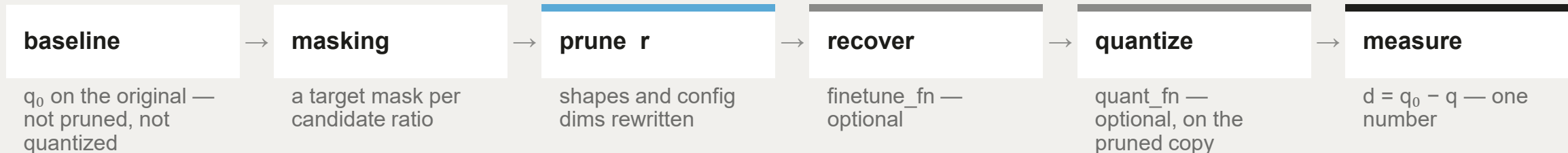


→ The pruning budget adapts to the threshold, removing only necessary components. Fine-tuning may be applied when needed. 修剪机制预算依据预设阈值动态调整，仅剔除冗余组件；必要时可实施精细化微调。

Combined compression with quantization

Respect the AMCT structure – quantization budget first, prune budget later.

CALL FLOW · 调用流程



JOINT BUDGET · 共同预算

the caller states one tolerance τ — it is never divided between the two stages

τ · one budget for the whole pipeline

without quant_fn — τ bounds the pruning loss alone; the quantization loss arrives afterwards, unbudgeted

d · pruning loss

unused

with quant_fn — the copy is quantized before it is measured, so both losses land in the same d

d · pruning loss + quantization loss

unused

So, d can only grow, so the accepted ratio can only shrink — quant_fn makes the search prune less, never more

why this order

pruning rewrites shapes, so quantization must be respected before pruning

→ Respects the quantization budget and sets a conservative, stable pruning budget / 遵循量化预算，保守且稳定地设定剪枝预算。

Adoption : one call, one tolerance

Structural pruning enters to AMCT while respecting the existing structure

MECHANISM · 机制

WHAT PRUNING TOUCHES

embed
2048

→

× 20 layers

attention · untouched

router

256 experts, top-8

→

norm
RMS

→

lm_head
157,184 × 2,048

1

model structure

FFN · CNN · MOE

2

output space

lm_head rows

3

runtime execution

router eligibility

1

ships today

On going (page 14)

2

On going (page 14)

THE CALL

already in CANN/AMCT

new in this AMCT-Pruning

```
amct.quantize(model, cfg, data=calib)
amct.prune(model, cfg, data=calib)
accuracy_based_auto_prune(...)
prune_finetune(model, train_data, steps=300)
guard = calib_nll

tolerance=0.02
```

RESULTS · 结果

Model	What was cut	How much	Result
Dense-LM · WikiText	FFN intermediate dim	27% · best 61%	1% perplexity budget
granite-1b-a400m	whole MoE experts	17%	1% perplexity budget
MNIST CNN	conv2 output channels	30%	0.9869 → 0.9871, and no finetuning at all
OLMoE	the scoring criterion	—	perplexity 46.9 → 35.3
LLaDA2.1-flash	256 → 160 experts	37.5%	100 GB → 64 GB, and GSM8K 95.5 → 79.0% (Why? → Need new method for new architecture)

→ Designed to preserve the existing repository as much as possible; users only need to change a single keyword / 尽可能保持现有仓库不变，用户只需修改一个关键词即可。

Ongoing researches

What's next

下一步

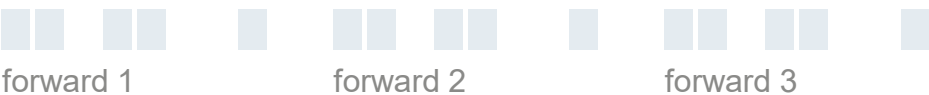
Pruning gave back memory but not always saves time. So what does give back time?

剪枝换回了显存，却没有换回时间。那么，什么才能换回时间？

→ The remainder of this talk focuses on our ongoing research on model acceleration / 本次分享的后半部分将聚焦于我们目前正在进行的模型加速研究。

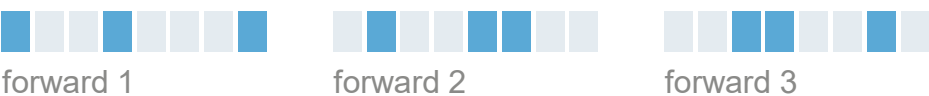
DELETE

the same experts are gone on every forward



SELECT

a different set each forward — none of them gone



IN THIS CHAPTER

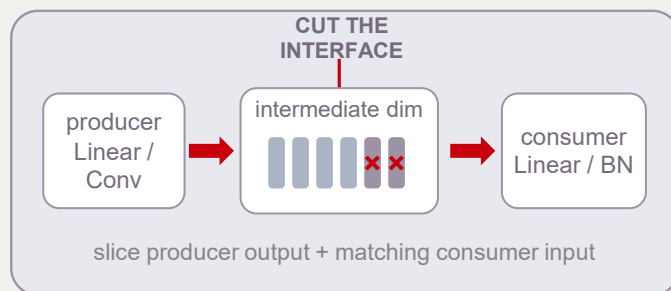
- Current Issues
- The decoder, and where the time is
- Per-problem and per-forward selection
- Three lenses, side by side

Current issues of pruning by inference steps

One pruning principle at **three** decision times

WHAT EACH LENS ACTUALLY PRUNES · 每种方法到底剪什么

BEFORE SERVING / 部署前

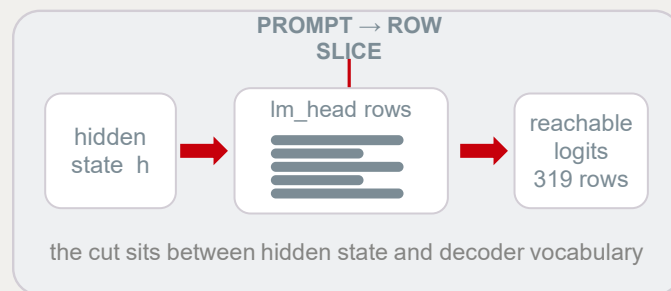


01 Model structure / 模型结构

selects weights · channels · experts
from calibration statistics

PERMANENT · one checkpoint

PER REQUEST / 每个请求

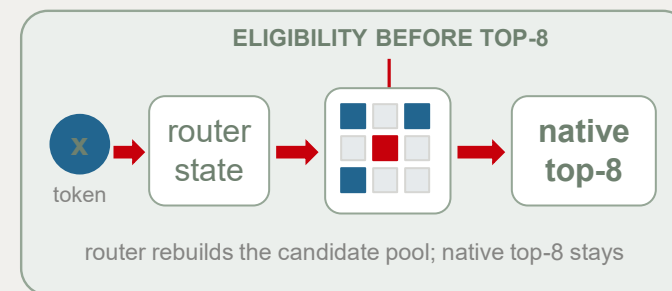


02 Output length pruning / 输出空间

selects reachable vocabulary
from prompt tokens and corresponding structure

REQUEST-LIVED · one problem

PER FORWARD / 每次前向



03 Active expert set / 活跃专家集

selects eligible experts
from router state

FORWARD-LIVED · one kernel call

Method depends on the part of the inference steps / 具体方法取决于推理过程的特定阶段

→ Each scope differs in what to prune, when to prune, and how long it stays. / 每个维度在修剪的内容、修剪的时间和保留时长上都有所不同。

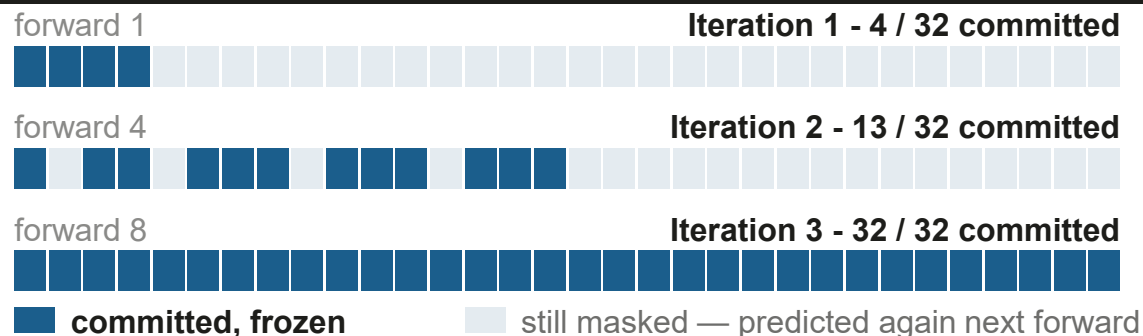
Case : Block diffusion decoder

下一步 · 13

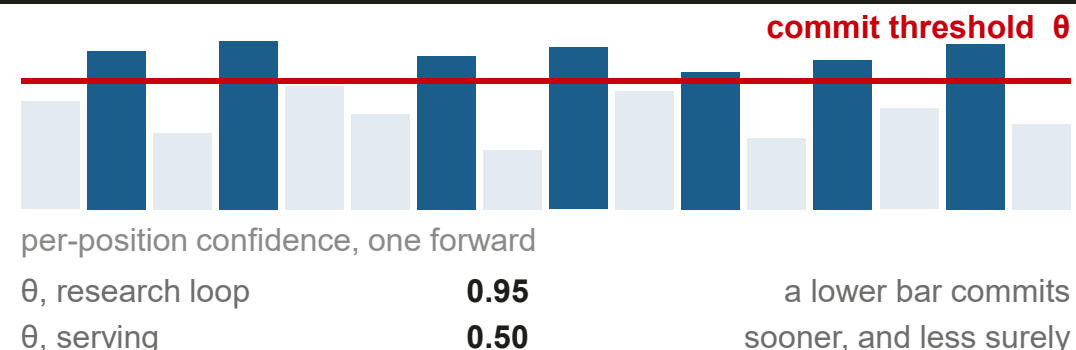
New model architecture may not work like expected.

MECHANISM · 机制

ONE BLOCK, 32 POSITIONS, MANY FORWARDS



A POSITION COMMITS WHEN IT CLEARS θ



WHERE THE TIME IS · 时间在哪里

$$\text{wall clock} = \text{forwards} \times \text{cost per forward}$$

LEVER A fewer forwards

commit more positions per pass, or need fewer passes

LEVER B a cheaper forward

restrict what a single forward is allowed to touch

they are **not independent**: restricting what a forward may touch makes it cheaper and makes commit takes longer

→ Modern models actively leverage test-time scaling and iterative computation, offering significant headroom for acceleration on our NPUs. However, their dynamic computation patterns make them difficult to optimize efficiently. / 现代模型通过积极引入测试时扩展与迭代计算机制，为神经处理单元（NPU）带来了显著的加速潜力。然而，此类模型所呈现的动态计算特性，亦给实现高效优化带来了严峻挑战。

MECHANISM · 机制

→ Both methods leave every weight in place and change only what one forward is allowed to touch. / 两种方法都不删除权重，仅调整单次前向传播过程中可访问的数据内容。

Three future lenses : side by side

结论 · 15

The same operation at three decision times

MECHANISM · 机制

WHEN IS THE COMPUTATION SELECTED?

later decision → more context, weaker repair

OFFLINE once per checkpoint

offline, before serving



sees: the calibration set
removes: weights, permanently
repair: one finetune pass
result: $\approx 0\% \rightarrow 27\%$, best 61%

PER PROBLEM once per request

output-space reduction



sees: this problem's prompt tokens
removes: output-head rows $\rightarrow 0.20\%$
repair: recalibrate the commit rule
result: a cheaper forward +50% but increased count

PER FORWARD every MoE layer

MMA 32+16



sees: this block's router state
removes: expert eligibility
repair: one scalar temperature
result: +50% tok/s but reduced acc, and deployment bottleneck

EVIDENCE · 证据

Property	Offline · AMCT	Per problem · output space	Per forward · MMA
Information available	calibration statistics only	the problem's own prompt tokens	this layer's router logits, this block
Granularity of the cut	one mask for every request	one vocabulary per request	one eligible set per layer per pass
Recovery mechanism	gradient descent	a commit rule refitted to the reduced support	a frozen scalar
What breaks	task accuracy	the commit threshold's scale + output space	per-token commit confidence



In the near future, optimizations from the three different scopes could work together synergistically, creating cumulative gains across the system. / 展望未来，源自三个维度的优化举措将形成协同效应，进而为系统整体带来累积性增益。

【有奖任务】AMCT Quantization & Pruning 样例贡献 🎁



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